

The Lagrange Loom

A Self-Weaving Orbital Tether Network for Propellantless Cargo Transfer in Cislunar Space

Concept Study LL-CS-001 · Off-World Civilisation Toolkit · July 2026

Classification: Speculative engineering, constrained to known physics

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Abstract

Every kilogram lifted between orbits today is held hostage by the rocket equation: to move mass, we burn mass, and most of what we launch is the fuel required to move the rest. This study describes the Lagrange Loom, a robotic spacecraft whose sole purpose is to make that arithmetic obsolete along fixed cislunar routes. The Loom manufactures and deploys ultra-high-strength tether segments, joining them into rotating and hanging tether structures anchored at gravitationally stable waypoints. Once woven, the network transfers momentum mechanically: a payload arriving from Earth donates its excess velocity to the web, and that same momentum is later spent flinging an outbound payload toward the Moon. No propellant changes hands. We present the operating principle, the orbital mechanics of momentum-exchange tethers, a materials assessment centred on carbon-nanotube and ultra-high-molecular-weight polyethylene fibres, an operations concept, and a staged technology roadmap. Nothing in the concept requires new physics; every subsystem descends from hardware that has already flown. What is missing is scale, and scale is precisely what a self-weaving machine is for.

1. Introduction: The Toll Road Problem

Imagine that every truck crossing a mountain pass had to carry, in its own cargo bed, the entire road it would drive on, and then throw the road away at the far side. That is, with only slight unfairness, how spaceflight works in 2026. A rocket bound for lunar orbit expends nearly all of its mass as propellant, and the propellant is gone forever. The road is rebuilt, at full price, for every single trip.

Civil engineers solved the terrestrial version of this problem millennia ago: they built the road once and let everyone use it. The Lagrange Loom is a proposal to do the same in space. It is not a ship that carries things. It is a machine that builds the road — a permanent mechanical infrastructure of tethers, hubs, and momentum banks strung between low Earth orbit (LEO) and the Earth–Moon Lagrange points, along which cargo climbs and swings the way a gondola crosses a valley.

The idea of exchanging momentum through long tethers is not new. Tsiolkovsky sketched towers to orbit in 1895; Moravec (1977) formalised the rotating skyhook; Hoyt and Forward showed in the 1990s that a staged Earth–Moon tether system could move cargo with essentially zero net propellant (Hoyt and Forward, 1997). Flight experiments — Gemini XI's spun tether (1966), the 19.7 km deployment of SEDS-1 (1993), TSS-1R's demonstration of kilometre-scale electrodynamic current (1996), and the student-built YES2 deorbit tether (2007) (Kruijff and van der Heide, 2009) — have confirmed the mechanics piece by piece. What has never existed is a machine that can manufacture, splice, inspect, and repair tether at orbital scale, autonomously, for years. That machine is the subject of this paper.

2. Concept Description

The Lagrange Loom is a 12-tonne autonomous spacecraft built around a central spinneret: a fibre-processing plant that accepts spooled precursor filament launched cheaply as dense cargo, and extrudes finished, braided, redundantly-interlinked tether ribbon. The braiding geometry follows the Hoytether architecture (Hoyt and Forward, 1997), a multi-line lattice in which any single severed strand — a micrometeoroid strike, say — sheds its load to neighbours instead of parting the whole line. The Loom does not merely deploy tether; it weaves structure, which is why the textile metaphor is engineering rather than poetry.

A completed network consists of three element types. Anchor hubs are ballast masses (spent upper stages, lunar regolith bags, retired satellites) parked at stable or quasi-stable locations — notably Earth–Moon L1 and L2, where a small station-keeping budget buys a fixed address in the sky. Transfer tethers are rotating momentum-exchange lines, tens to hundreds of kilometres long, whose tips periodically dip to low relative velocity to catch and release payloads (Hoyt, 2005). Climber lines are non-rotating hanging tethers along which slow cargo pods winch themselves, trading electrical energy for altitude at a fraction of rocket cost. The Loom crawls this growing web like its namesake spider, extending lines, splicing junctions, and replacing aged segments.

3. Physics of the Web

The central transaction of the network is the momentum exchange. A rotating tether of length L spinning with tip speed v behaves, at the moment of payload capture, like a moving handhold: a payload arriving with the right relative velocity is grappled at the tip, swings half a rotation, and is released with a velocity change of up to $2v$ relative to the tether's centre of mass — all without burning anything (Moravec, 1977; Hoyt, 2005). The tether itself recoils, its orbit dropping slightly; it has banked the payload's momentum. That bank balance can be restored three ways: by catching traffic moving in the opposite direction (the elegant solution — inbound lunar cargo pays for outbound Earth cargo), by slow electrodynamic thrusting against Earth's magnetic field using solar power and no propellant (Sanmartín, Martínez-Sánchez and Ahedo, 1993), or by ion-engine trim as a last resort.

The governing constraint is the tip acceleration. A tether of tip speed v and length L imposes centripetal acceleration $v^2/(L/2)$ on its own material, which sets the mass ratio between tether and payload through the characteristic velocity of the fibre — the speed at which a material can spin its own weight. For Zylon and UHMWPE fibres flying today, characteristic velocities exceed 2.5 km/s, making 1–1.5 km/s tip-speed tethers practical with tether-to-payload mass ratios near 20:1 (Hoyt and Forward, 1997). Laboratory carbon nanotube yarn pushes the theoretical figure several times higher, collapsing that ratio toward 3:1. The Loom is deliberately fibre-agnostic: it weaves whatever the best current spool is, and reweaves as materials improve. Infrastructure that can moul.

Placement at the Lagrange points is what turns isolated tethers into a network. L1, hovering permanently between Earth and Moon, is the natural interchange: a payload released from an Earth-side rotating tether coasts uphill to L1, is caught by the hub, and is handed to a Moon-side line that lowers it toward lunar orbit. Because L1 and L2 are saddle points rather than wells, hubs there require modest station-keeping — of order 10 m/s per year (Farquhar, 1970) — which the network's electrodynamic segments can fund. The whole system behaves less like a fleet of vehicles than like a conveyor: mass flows down the potential gradient in one direction pay the toll for mass flowing up it in the other.

4. The Machine Itself

The Loom's anatomy follows its verbs: spin, splice, crawl, inspect. The spinneret bay houses drum-fed braiders adapted from terrestrial rope-walk machinery, vacuum-rated and thermally managed so that polymer fibre never sees the resonant heating that degrades it. Twin articulated weaving arms handle splicing: each junction in the web is a mechanically interlocked, adhesive-free knot geometry chosen so that inspection is visual and repair is reversal. A crawler drive of paired pinch-rollers lets the Loom translate along existing lines at walking pace without abrading them. A fibre-optic interrogator threads a sensing filament through every ribbon it weaves, so the finished web reports its own strain map continuously — the network is its own structural health monitor.

Autonomy is not optional. The Loom operates seconds to minutes of light-lag from its operators and performs manipulation tasks (catch, splice, tension) whose timing tolerances are far tighter than a human teleoperation loop. Its AI foreman descends directly from the rendezvous-and-capture autonomy already proven by uncrewed cargo vehicles, extended with a weaving-plan solver that treats the web as a growing graph and schedules work orders against traffic forecasts. Crucially, the machine is conservative by design: every operation is reversible, every splice is redundant, and the failure mode of the network is graceful sag, not catastrophic parting.

5. Operations Concept

Phase one is a single momentum-exchange tether in LEO — a toll booth, not yet a road. It catches suborbital or barely-orbital payloads lofted by small launchers and throws them to geostationary transfer orbit, immediately multiplying the effective capacity of cheap rockets. Phase two extends a hanging climber line and plants the first ballast hub at L1, opening a slow-freight lane for non-urgent cargo: propellant, water, structural stock. Phase three closes the loop with a lunar-side rotating tether whose tip skims low lunar orbit, at which point a bag of Moon regolith bound for Earth-orbit industry literally pays the momentum bill for a habitat module bound for the lunar surface. The road begins collecting tolls in both directions.

Parameter	Phase 1 (LEO)	Phase 3 (Full Network)
Tether length	100 km rotating	3,500 km cumulative web
Tip speed	1.2 km/s	1.5 km/s (Earth-side)
Payload class	500 kg	10,000 kg
Propellant per transfer	< 5% of rocket equivalent	~0 (momentum-balanced)
Fibre	UHMWPE / Zylon Hoytether	CNT-hybrid ribbon
Loom units deployed	1	6 (weaving + maintenance)

Table 1. Representative parameters for the staged build-out.

6. Honest Difficulties

Three problems dominate. Debris and micrometeoroids will cut strands constantly; the Hoytether lattice converts this from a mission-ending event into a maintenance queue, but the maintenance queue is real, and it is why the Loom must live on the web rather than visit it (Hoyt and Forward, 1997). Capture dynamics are unforgiving: a tip catch is a rendezvous with a target moving on a cycloid, demanding

guidance in the class of — but beyond — existing autonomous docking. And traffic imbalance is an economic risk masquerading as physics: if cargo flows only outward, the momentum bank drains and must be refilled electrodynamically, slowly. The network wants two-way trade the way a railway wants return freight; its business case and its orbital mechanics are the same curve.

7. Pathway to Reality

Every subsystem sits at technology readiness levels between laboratory demonstration and flight heritage. Kilometre-class tether deployment has flown repeatedly (Kruijff and van der Heide, 2009); electrodynamic reboost was demonstrated in principle on TSS-1R and studied in depth under NASA's MXER tether programme (Hoyt, 2005; Sanmartín, Martínez-Sánchez and Ahedo, 1993); autonomous capture flies weekly to the ISS; high-tenacity fibre is an industrial commodity, and CNT yarn is leaving the lab. The genuinely novel item — orbital fibre processing and splicing — is a robotics problem, not a physics problem, and it is exactly the kind of problem that improves at the rate of terrestrial automation. A defensible guess places a Phase 1 toll booth within twenty years of a funded start, with the full Loom network a mid-century project: ambitious, expensive, and entirely legal under the laws of physics.

8. Conclusion

Rockets will always be how we leave a planet's surface; nothing here retires them. But between orbits, in the calm where there is no atmosphere to fight and no gravity well to climb quickly, transport can stop being ballistics and start being infrastructure. The Lagrange Loom is a bet that the second space age will be built the way every other frontier was tamed — not with better wagons, but with roads. The spider, patient and propellantless, just has to keep weaving.

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Harvard referencing style (Cite Them Right).

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